# Manthena Asmitha

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2028 | 7.28 |
| Aditya Junior College West Godavari | Class XII (Board of Intermediate Education Andhra Pradesh) | 2024 | 97.80% |
| Aditya Public School West Godavari Palakollu | Class X (Board of Secondary Education Andhra Pradesh) | 2022 | 95.50% |

## SCHOLASTIC ACHIEVEMENTS

- Secured **100% in Mathematics, Physics & Chemistry** in Class XI BIEAP Board.
- Secured **100% in Mathematics** in Class XII BIEAP Board.

## POSITIONS OF RESPONSIBILITY

**Infrastructure Coordinator -- ELAN & Vision**

- Coordinated with vendors and internal teams to **communicate requirements, obtain quotations, and arrange necessary resources** and identified, shortlisted, and **contacted vendors** based on event-specific requirements ,coordinating with multiple stakeholders .
- Handled **venue sourcing and booking**, aligning logistical requirements with event schedules and operational constraints.
- Coordinated **infrastructure and logistics requirements** for events, ensuring timely availability of venues, equipment, and other resources.

## SKILLS

**Technical Skills:**

Python, MATLAB, ABAQUS, Finite Element Analysis (FEA), Structural Analysis, Engineering Data Analysis, Statistical Analysis & Data Visualization

**Programming languages:** C, C++, Python

**CAD & Design:** Solid edge, AutoCAD

## RELEVANT COURSES

**Mechanical:** Solid and fluid mechanics, Engineering drawing, Digital fabrication, Manufacturing science, Engineering mechanics, Heat and mass transfer, Kinematics and dynamics of machinery,Thermodynamics

**Mathematics:** Calculus, Differential equations, Probability, Statistics, Linear Algebra

## PROJECTS

**Structural Analysis Project -- Truss Structures**

- Performed **analytical analysis of truss structures** to determine member forces and stresses under specified

loading and boundary conditions.
- Developed and evaluated truss models using **FEA/software simulations**, comparing numerical results with analytical calculations.
- Investigated discrepancies between analytical and simulation results and **validated the model** by checking assumptions, boundary conditions and member behaviour.
- Used a **data-driven comparison** of analytical and computational results to assess the accuracy and reliability of the structural model.

**Kinematics & Dynamics of Machinery** | *GitHub*

- Conducted **kinematic simulation and analysis of four-bar and slider-crank mechanisms**, including their various inversions, using **Python** to study displacement, velocity, acceleration, and motion characteristics.
- Developed **Python-based computational models** to analyze mechanism motion and visualize the kinematic behavior of individual links and joints.
- Performed **mechanism synthesis** involving **revolute (R) and prismatic (P) joints** to achieve prescribed motion and positional requirements of the moving body.
- Used Python for **numerical calculations, trajectory generation, and graphical visualization** to validate the performance and motion of synthesized mechanisms.

**3D Modelling and Additive manufacturing**

- Designed and developed a **3D CAD model** using **Solid Edge**, incorporating accurate dimensions, geometric constraints, and design specifications to ensure dimensional accuracy and manufacturability.
- Applied **parametric modeling and design constraints** to create a robust and easily modifiable CAD model while maintaining proper geometric relationships between features.
- Performed **slicing and print setup**, configuring parameters such as layer height, infill, supports, and print orientation to achieve a balance between **print quality, strength, material usage, and printing time**.

**Statistics & Data Visualization Project | Python**

- Analyzed **MS Dhoni's ODI career data** to identify performance patterns across opponents, scoring ranges, and batting positions.
- Used **Python and statistical visualization techniques** including bar charts, frequency distributions, histograms, and box plots to transform raw data into interpretable insights.
- Compared performance distributions across batting positions using **median, spread, frequency and outlier analysis**, identifying differences in consistency and scoring patterns.
- Synthesized quantitative findings into **clear conclusions and visual representations**, demonstrating data-driven reasoning and communication.

## EXPERIENCE

**Engineering Intern -- PSuits**

- Conducted **Finite Element Analysis (FEA)** to evaluate engineering applications and understand performance under different conditions.
- Researched and **collected technical data on multiple adhesive materials**, comparing their properties against application-specific requirements.
- Synthesised technical information from multiple sources into **clear, data-driven conclusions**, supporting material-selection decisions.